Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 21: Sizes of Infinity

Cardinality, Countability, and Cantor's Argument

Chartrand & Zhang, Section 5.5

Name: _____ Date: _____

Where we are — and a note on pace

This closes Chapter 5. We are covering Section 5.5 in one period rather than three, so this is the highlights: the definition, two examples, and one famous proof. The section has more, and it is worth reading.
Everything here rests on **bijections**, which you built last week. The new idea is what we use them *for*.

By the end of today you should be able to

1. Say what it means for two sets to have the same cardinality.
2. Explain why $\mathbb{Z}$ is denumerable.
3. Sketch why $[0, 1]$ is not.

1. Same size means a bijection

For finite sets, $|A| = |B|$ means the same number of elements — and equivalently, that the elements can be **paired off**. Pairing off is the idea that survives to infinite sets.

Definition

Two sets A and B have the **same cardinality**, written $|A| = |B|$, if there exists a bijective function from A to B.

Denumerable

A set A is **denumerable** (or **countably infinite**) if $|A| = |\mathbb{N}|$ — that is, if its elements can be listed as $a_1, a_2, a_3, \ldots$ with every element appearing exactly once.

A listing *is* a bijection from $\mathbb{N}$: position n goes to a_n.

2. $\mathbb{Z}$ is denumerable

$\mathbb{Z}$ looks twice as big as $\mathbb{N}$ — it runs both ways. But we only need a listing.

$$0,\ 1,\ -1,\ 2,\ -2,\ 3,\ -3,\ \ldots$$

Every integer appears exactly once, so this is a bijection $\mathbb{N} \to \mathbb{Z}$. Hence $|\mathbb{Z}| = |\mathbb{N}|$.

This is the strangeness of infinity, not a trick

$\mathbb{N}$ is a *proper subset* of $\mathbb{Z}$, and they have the *same* cardinality.
You already saw the crack this comes through. In Practice Set 19 you proved that for a **finite** set, a one-to-one function $A \to A$ must be onto — and the proof leaned on finiteness. Lecture 19's $f(n) = 2n$ is one-to-one from $\mathbb{Z}$ into a proper subset of itself. For infinite sets, part of a thing can be the same size as the thing.

Also denumerable (stated, not proved today): the rationals $\mathbb{Q}$, every infinite subset of a denumerable set, and the union of two disjoint denumerable sets.

That $\mathbb{Q}$ is denumerable is genuinely surprising — between any two rationals lies another, yet they can still be put in a list.

3. Countable and uncountable

Definition 5.80

A set is **countable** if it is finite or denumerable. A set that is not countable is **uncountable**.

So $\{1, 2, 3\}$, $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ are all countable. The question is whether anything is not.

4. Cantor's diagonal argument

Theorem 5.79

The interval $[0, 1]$ is uncountable.

Proof sketch (by contradiction). Suppose $[0, 1]$ *were* denumerable. Then every number in it appears in some list:

$$a_1 = 0.a_{11}a_{12}a_{13}\cdots$$
$$a_2 = 0.a_{21}a_{22}a_{23}\cdots$$
$$a_3 = 0.a_{31}a_{32}a_{33}\cdots$$

Now build a new number $b = 0.b_1b_2b_3\cdots$ by walking down the **diagonal** and changing each digit:

$$b_n = \begin{cases} 2 & \text{if } a_{nn} = 1 \\ 1 & \text{if } a_{nn} \neq 1 \end{cases}$$

Try it. If $a_1 = 0.111\ldots$, $a_2 = 0.02999\ldots$, $a_3 = 0.141414\ldots$, $a_4 = 0.135167\ldots$, then the diagonal digits are $1, 2, 1, 1$, so

$$b = 0.2122\cdots$$

Now $b \in [0,1]$, but $b \neq a_1$ because they differ in the first digit, $b \neq a_2$ because they differ in the second, and in general $b \neq a_n$ for every n.

So b is missing from a list that was supposed to contain everything — a contradiction. ■

Two consequences

Since $[0,1] \subseteq \mathbb{R}$ and any set containing an uncountable subset is uncountable (Theorem 5.81), $\mathbb{R}$ **is uncountable** (Corollary 5.82).
And since $\mathbb{Q}$ is countable while $\mathbb{R}$ is not, the irrational numbers are uncountable (Theorem 5.85). In a precise sense, almost every real number is irrational.

5. There is no largest set

Theorem 5.86

For every set A, $\quad |A| < |\mathcal{P}(A)|$.

Cantor's proof assumes a bijection $g : A \to \mathcal{P}(A)$ and then builds the set

$$A_g = \{a \in A : a \notin g(a)\}$$

— the elements *not* contained in the set they are assigned to. If $A_g = g(a_0)$ for some a_0, then asking whether $a_0 \in A_g$ gives a contradiction either way.

Day 1, doing real work

That last step is the shape of "This sentence is false" — the Liar sentence from Lecture 1, which we classified as *not a statement* and moved on from.
Here the same self-reference is not a curiosity. It proves there is **no largest cardinality**: from any set, the power set is strictly bigger, forever. It is also the shape of the halting-problem argument you will meet in theory of computation.

Exit ticket

1. What must you exhibit to show two sets have the same cardinality?

2. Is $\mathbb{Q}$ countable or uncountable? Is $\mathbb{R}$?

Before Wednesday

Read Section 3.2, "Direct Proof," pp. 87–96. We begin proof methods — the core of the course — and everything from here builds on it.

Self-check (not collected). Section 5.5, Exercises 1, 3, and 9 — odd-numbered, answers in the back.

Coming up

Homework 8 is due Friday, October 9, 11:59 p.m.
Midterm: Wednesday, October 14, covering Chapters 1, 2, and 5. Monday, October 12 is a review session.

Questions I still have